Examining the Long Term Effects of Redlining in the United States and the Phoenix Metropolitan Area

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1 Abstract/Executive Summary

This project investigates the lingering effects of redlining, a discriminatory practice formalized by the Home Owners' Loan Corporation (HOLC) during the 1930s. Our data analysis covers demographic shifts, educational proficiency, environmental justice, and socioeconomic disparities in Phoenix, Arizona. Employing a variety of methods including statistical analysis and machine learning models, this project explores the correlations between redlined areas and various aspects of life and infrastructure.

There is a strong correlation between redlined areas and a lowered standard of living. There are disparities in racial distributions across all HOLC rated areas. 67.2 percent of the population of HOLC A areas were white and 59 percent of the population of HOLC D areas were black and Hispanic. When comparing distric finance data, districts in HOLC A areas had the highest average revenues, local revenues, and expenditures. Environmental data analysis reveals that areas with a lower grade given by the HOLC face have higher exposure to pollutants and rates of asthma, affecting the health of the residents in these areas significantly. When training a random forest model the predicted the HOLC grade using a comprehensive dataset containing all of our observed variables, we were able to predict the HOLC zone with an accuracy of 96.57 percent.

2 Introduction/Background

Between 1934 and 1940, the Home Owner's Loan Corporation (HOLC), a U.S. Federal Agency created as part of the New Deal, assessed mortgage investment and lending risks of neighborhoods across the United States and assigned areas grades based on their perceived risk. These grades were reflected on what the HOLC called residential security maps, which assigned colors to areas based on their grades. Areas that posed a high risk to lenders were given a grade of *D* or *Hazardous* and were assigned to the color red respectively. The areas that received a *D* grade or were deemed *Hazardous* were predominantly black and brown neighborhoods and the HOLC discouraged borrowers from lending to those in D neighborhoods. These maps allowed the HOLC, lenders, and other institutions to categorically deny loans to borrowers indirectly on the basis of class and race [7, 3, 4].

The HOLC used the neighborhood grades as a way of discrimination to those in black and brown communities. By deeming black and brown neighborhoods as class D neighborhoods, they were legally allowed to discourage lending to those in D neighborhoods, who happened to be people of color. The practice of denying people access to loans and credit on the basis of class and race became known as redlining. This practice limited access to home ownership in

areas with higher minority populations and reinforced racial segregation by denying financial opportunities on the basis of class and race [8].

The impact of redlining persists in various aspects of life in the United States, influencing factors such as the racial composition of communities, education, health, and socioeconomic development.

2.1 Lingering Population Effects of Redlining in the United States

While the Fair Housing Act of 1968 outlawed racially motivated redlining, redlining has shaped the modern landscape of housing [8].

In 2022, FiveThirtyEight, an organization focused on opinion poll analysis, sought to find the lasting impacts of redlining in a report called The Lasting Legacy of Redlining. The report found that when comparing the maps created by HOLC in the 1930's with the 2020 census, the areas that were deemed "hazardous" by the HOLC, or formally redlined zones, are still disproportionately Black, Latino, or Asian compared with their surrounding metropolitan area and that the areas that were considered "best" for lending are predominantly white. In the city of Chicago, there are 90 percent more Black residents living in formerly redlined areas than in surrounding areas. In 50 cities in the United States that had large Latino populations, FiveThirtyEight found that over 72 percent of Latinos were living in formerly redlined areas [2].

2.2 Redlining Effects on Education in the United States

In a study conducted by Annenberg Institute at Brown University, the authors sought to empirically describe the legacy of redlining on school funding, diversity, and performance [6]. In the study, they leveraged spatial analysis of 144 HOLC security maps to understand how the HOLC maps relate to current patterns of school funding, diversity, and performance. They combined HOLC data with geospatial district and school-level data from the National Center for Education Statistics, school-level finance data from the Edunomics Lab at Georgetown University which produced the National Education Resource Database on Schools, and Stanford Education Archive data for school performance analysis to find the lingering effects of redlining on education. In their analysis, they found that average total per-pupil revenues are lower for districts mapped to HOLC D grades than those mapped to other grades on a federal, state, and local level. Additionally, the student of populations of districts mapped to HOLC D grades were predominately black and brown and districts mapped to HOLC A and B grades were predominately white. Finally, they found that students in HOLC A, B, and C grades performed better in mathematics and ELA assessments than their HOLC D grade counterparts [6]. This study reveals the lingering effects of redlining on education.

2.3 Redlining Effects on Health

In a study conducted in 2022, researchers combined the HOLC security maps with data from the Center for Air, Climate, and Energy Solutions to see if there was a relationship between present-day air pollution disparities in U.S. cities and historical redlining [5]. The study concluded that people of color experience higher-than-average nitrogen dioxide and pm 2.5 levels and are over-represented within HOLC C and D grade neighborhoods [5]. While exposure to harmful air-pollutants is not an underlying measure of health, this paper gives insights on how people residing in HOLC D grade areas are disproportionately exposed to conditions that affect health.

Similarly, a study found that when combining HOLC security maps of Milwaukee with infant mortality rate data collected from the City of Milwaukee and mental and physical health data from the Centers for Disease Control and Prevention's 500 Cities Project, there is enough

evidence to suggest that prevalence of poor physical and mental health and infant mortality rates were correlated with areas with HOLC D graded areas [7]. Again, while the prevalence of poor physical and mental health are not underlying measures of health, this report highlights how people who live in formerly redlined areas are at risk of poor health.

2.4 Redlining and its Effects in Phoenix, Arizona

While outlawed over five decades ago, redlining effects continue to reverberate through the city's social and economic landscape, particularly for Black, Indigenous, and Latinx communities. The HOLC perpetuated these inequities throughout the USA, including in Phoenix, Arizona. Due to actions by the federal government, racially restrictive covenants were present in Arizona as early as 1913 and lasted until the mid-1950's. Racial covenants further cemented segregation by barring minorities from purchasing homes in certain areas [1]..

In the 1950s, Native American families were urged to the cities and away from their reservations, many ending up in lower-class neighborhoods that had high populations of Black and Hispanic residents. Many of these urban, low-income areas had limited access to public services due to discriminatory government policies [1]..

These areas were then subject to 'revitalization' and highway construction that displaced over 300,000 families. Most of these projects occurred in these historically redlined areas, destroying Black and Latino neighborhoods [1].

The Phoenix metropolitan area faces rapid growth, but for Black, Indigenous, and Latinx residents, this progress often masks persistent discrimination. This discrimination extends to education, employment, and most notably, housing, a direct consequence of historical redlining practices. Redlining effectively prevented many families from purchasing properties on the north side, Maricopa County. Those discriminatory policies forced them to be relegated to the south of Van Buren Avenue. Neighborhoods in this area were in worse conditions than those in the north - filled with environmental and toxic hazards, factories, freeways, and noise pollution. Historical redlining also has future implications that can result in gentrification and displacement of existing residents and businesses [9]. The effects of historical redlining still prevail, which we will examine in our project, we will delve deeper into the enduring effects of redlining, and examine its impact.

2.5 Environmental Injustice in Phoenix, Arizona as the Result of Redlining

South Phoenix is marred by industry and hazardous waste that deeply affects its residents. This is an environmental justice issue, and this project will look into correlations between air pollution, disease rates, and other factors that are a result of environmental stress.

The east-west railroad line that was created by redlining served as a demarcation line that divided races and social classes, setting South Phoenix apart. In the 1950s, 75 percent of Phoenix's factories were located close to the railroad. Then, the railroad brought migrant workers to Phoenix. Now, they bring hazardous waste that is processed and stored in South and West Phoenix. Due to these facilities as well as storage yards, small manufacturers, auto body repair shops, major freeways cutting through the area, and Phoenix's Sky Harbor International Airport, air quality is very bad in South Central Phoenix. Levels of particulate matter regularly ranged from 62-87 above the health standard [11].

2.6 Socioeconomic Consequences of Redlining in Phoenix, Arizona

The socioeconomic consequences of redlining in Phoenix have had an enduring impact on the city's landscape, affecting income levels, unemployment rates, poverty rates, access to opportu-

nities, and the ability for wealth accumulation among marginalized communities. This practice, rooted in institutionalized racism, has not only segregated neighborhoods but also perpetuated economic disparities that linger across generations. In the 1930s, the federal government designated south Phoenix as a hazardous area, denying residents access to loans and mortgages. This practice, along with discriminatory restrictions, contributed to low homeownership rates among Black and Latino residents. Despite the illegality of redlining over five decades ago, its repercussions echo in the present, manifesting in heightened foreclosure rates during economic downturns and a recent surge in rents and home prices that increasingly displace long-time residents. The revitalization efforts and infrastructural investments, while beneficial, often fall short of addressing the systemic inequities at play, leaving the community in a continuous struggle for equity and inclusion [10].

3 Methods

3.1 Racial Makeup in Different HOLC Zones

In order to determine the impact that redlining has had on the racial makeup of formerly redlined zones, we used FiveThirtyEight's dataset called 'metro-grades', which merged HOLC Maps reproduced by Mapping Inequality and the 2020 Census. This dataset contains the population estimate of all HOLC-grade areas. We then found the racial distributions of each HOLC zone.

3.2 Comparing School District Funding Across HOLC Zones

In order to analyze how education has been impacted by the practice of redlining, we compared the financial data of school districts in HOLC rated areas. To do so, we downloaded data from the National Center of Education Statistics provides the geodata for every school on a school and district level. We then spatially joined the district geodata and geocodes with HOLC Maps, to see which schools were in HOLC rated. The district geodata used the U.S. Census Tiger/Line to get geometries for each district.

After determining which school districts were in HOLC areas, we merged the geodata with F-33 survey data: an annual survey in which schools report how they spend and receive money. From here, we were able to determine how the total revenues, expenditures, and local revenues compared across different HOLC areas. Again, F-33 survey data was provided by the National Center for Education Statistics.

3.3 Environmental Justice

3.3.1 Data Preparation and Processing

In order to examine health issues that people in the HOLC neighborhoods face, we needed to look at the data based on census tracts. The dataset used is from Diversity Data Kids, an organization that uses research and data to enhance racial and justice equity. This dataset includes neighborhood grades by the HOLC mapped to 2010 US Census Tracts.

For the purposes of this analysis, the census tract's HOLC grade will be assigned based on class1 - which indicates the HOLC grade (A, B, C, or D) that covers the largest proportion of the census tract's area. It is the primary indicator of the HOLC rating for a census tract.

Combining this dataset with a shapefile showed the census tracts with a map of Arizona roads. This was combined with a dataset from the CDC for asthma in 2017 as well as PM2.5 concentrations, based on the 2010 census tracts.

3.3.2 Exploratory Data Analysis(EDA)

This project uses correlation analysis to examine the relationships between air quality, asthma prevalence, and historical housing grades. This is visualized through geospatial mapping, highlighting the geographic distributions of these issues and their correlations with redlined areas, scatter plots to show the distribution, and bar graphs to show weighted averages.

3.3.3 Statistical Analysis

A correlation matrix was used to investigate the relationship between air quality, asthma prevalence, and historical housing grades.

3.4 Economic Impact across different grades

Data for this study was aggregated from the National Historical Geographic Information System (NHGIS), U.S. Census datasets, and various publicly available datasets focused on county-specific socioeconomic data. To ensure accurate comparisons, these datasets were merged using geographical information, allowing for a seamless integration of demographic and economic data from different sources. Spatial joins were executed to overlay contemporary socioeconomic data onto historical HOLC maps. This critical step bridged past and present data, anchoring our analysis in a spatial context.

3.4.1 Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) was conducted to assess socioeconomic trends within HOLC-graded areas, focusing on educational attainment, median household income, and poverty rates. We analyzed educational levels, distinguishing between individuals with at least a bachelor's degree and those who completed high school, to identify disparities across different HOLC grades. Median household incomes, adjusted for current-year dollars using the latest census data, were mapped to these areas to evaluate economic distribution. Additionally, poverty rates were examined by integrating census data with HOLC geographical boundaries to understand how economic hardship correlates with historical redlining designations.

3.5 Regression Analysis

To assess the influence of various socioeconomic indicators on poverty rates, we employed regression analysis. Two models were developed: one incorporating median income and HOLC grades, and another excluding these factors to gauge their impact. The regression models helped quantify how different variables such as education levels and income disparities predict changes in poverty rates within the studied counties.

3.5.1 Analysis of Variance (ANOVA) and Post-hoc Analysis

ANOVA was utilized to examine differences in socioeconomic factors such as median income, education levels, and Gini indices across different HOLC grades. This statistical approach allowed us to determine whether the means of these indicators varied significantly by grade, providing insights into the economic impacts of redlining. Following significant findings from the ANOVA, Tukey's Honestly Significant Difference (HSD) test was conducted as a post-hoc analysis to further explore which specific HOLC grades differed from each other. This test compared all possible pairs of means, offering a detailed breakdown of pairwise comparisons and helping identify specific patterns of disparity among the grades.

3.5.2 Heatmap Visualization

To visually represent correlations among multiple socioeconomic variables, a heatmap was created using correlation coefficients. This visualization highlighted the strength and direction of relationships between variables such as income, poverty rates, and educational attainment. Heatmaps served as an intuitive tool for quickly identifying highly correlated variables, aiding in the interpretation of complex data relationships.

3.6 Predicting an Area's Rating

The literature associated with the lingering effects of redlining in the United States suggests that while the practice of redlining has been formally outlawed, there is evidence to suggest that the effects of redlining are still prevalent to this day. Our initial findings reveal disparities in population, school finance, income, and health amongst former HOLC A and HOLC D areas. Given the literature associated with the effects of redlining and our initial findings, we developed a random forest model to regrade areas based on current population, school finance, income, and air quality data in order to understand the lingering-effects of redlining. The model used data from all of the sources outlined above as features and an area's rating as the target variable. HOLC zones were encoded numerically, with HOLC A being 1, and HOLC D being 4.

4 Results

4.1 Racial Makeup in Different HOLC Zones

After examining the racial make up of all formerly graded areas, we found that as of 2020, HOLC A areas are predominately white, with white people making up 67.2 percent of the population in HOLC A areas. Comparatively, black and hispanic people combined for 56.9 percent of the population in HOLC D areas. This persisted in Phoenix, with white people making up for 67.3 percent of HOLC A areas and Hispanic people alone accounted for 60.7 percent of the HOLC D areas.

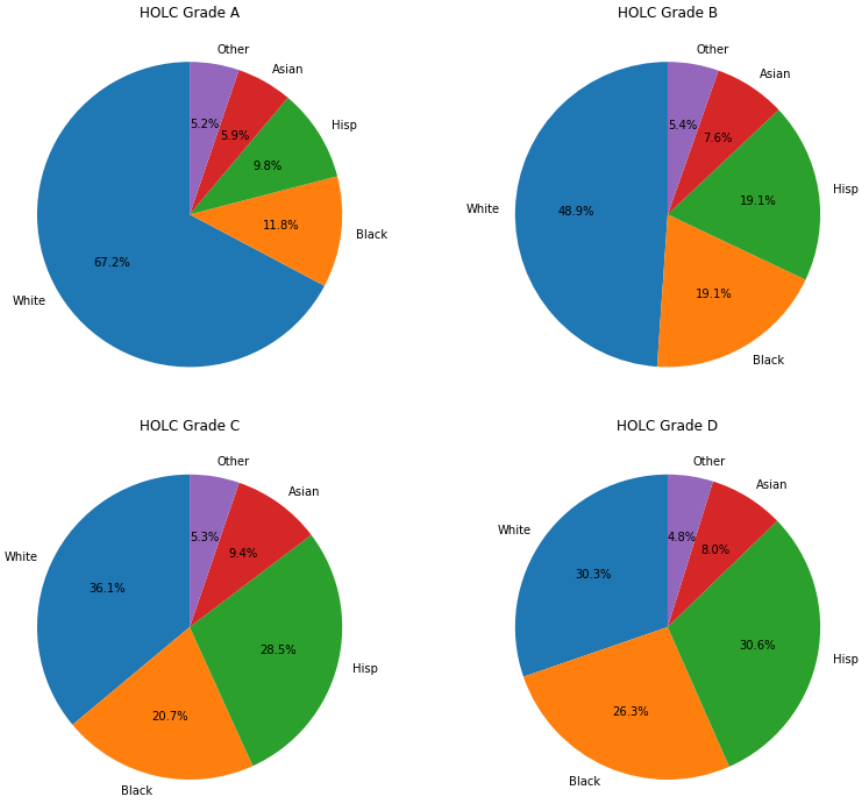

Figure 1: Racial Distribution of Population Across HOLC Zones in the United States

Figure 2: Racial Distribution of Population Across HOLC Zones in Arizona

The racial distributions of populations across the United States and Arizona according to the 2020 Census suggest that while redlining did not establish segregated communities, it continues to reinforce segregated communities. Like in the 1930s when the HOLC Residental Security Maps were created, white people tend live HOLC A rated areas were as black and brown people tend live in HOLC D areas.

4.2 Financial Disparities Among Districts in Different HOLC Areas

When comparing the financial data of districts across all HOLC Areas, we found that HOLC A areas had the highest average expenditures, revenues, and local revenues.

```
Average Expenditures for HOLC A: 65217288.461538464
Average Expenditures for HOLC B: 59000666.55686274
Average Expenditures for HOLC C: 55647182.22720898
Average Expenditures for HOLC D: 35817259.15993266
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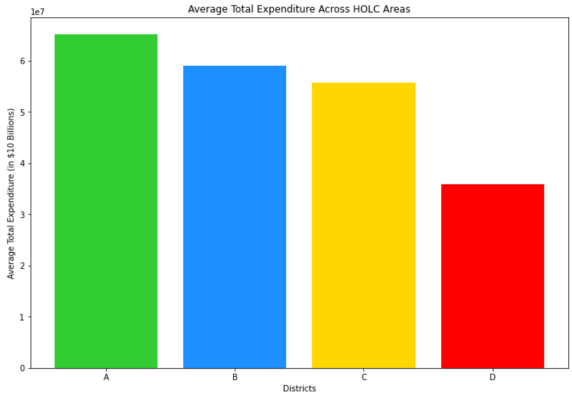

Figure 3: Average Total Expenditures Across HOLC Zones

```
Average Revenues for HOLC A: 64737826.92307692
Average Revenues for HOLC B: 58292176.360784315
Average Revenues for HOLC C: 58071098.07573633
Average Revenues for HOLC D: 33350203.604377106
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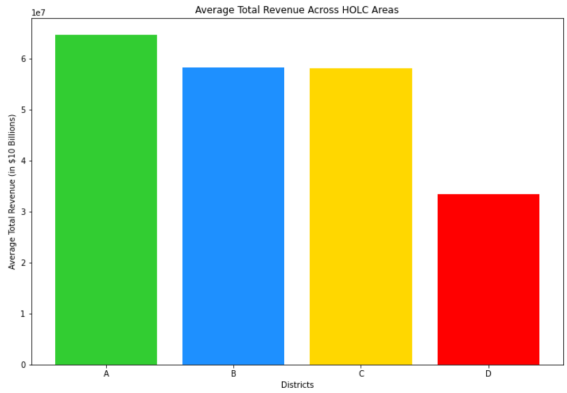

Figure 4: Average Total Revenues Across HOLC Zones

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Average Local Revenue for HOLC A: 33779596.15384615
Average Local Revenue for HOLC B: 28078019.498039216
Average Local Revenue for HOLC C: 23146291.62412342
Average Local Revenue for HOLC D: 13203388.789562289
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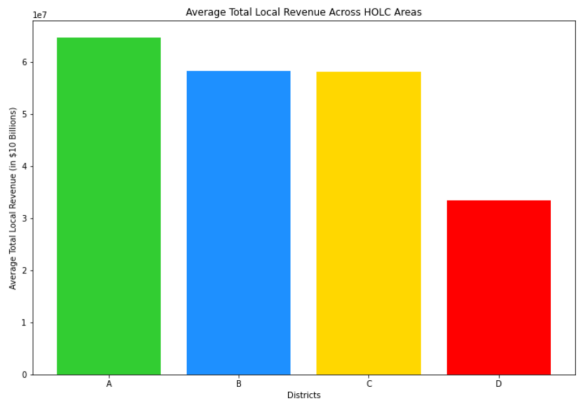
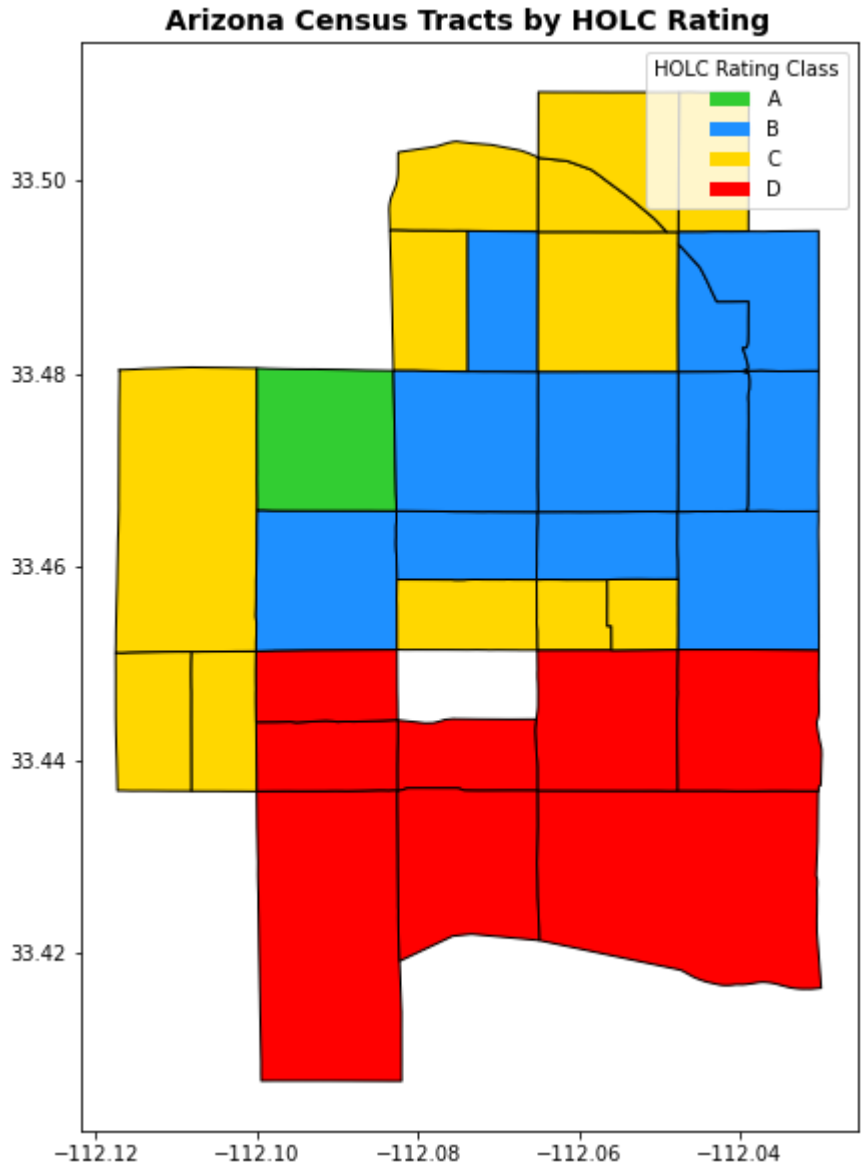


Figure 5: Average Total Local Revenue Across HOLC Zones

Districts in HOLC A zones had average expenditures, revenues, and local revenues at least 1.5 times greater than districts in HOLC D areas. This suggest that districts in HOLC D areas can lack the resources that districts in HOLC A area have, affecting the quality of education.

4.3 Environmental Justice

In order to examine health issues that people in the HOLC neighbhorhoods face, we needed to look at the data based on census tracts. The dataset used is from from Diversity Data Kids, an organization that uses research and data to enhance racial and justice equity. This dataset includes neighborhood grades by the HOLC mapped to 2010 US Census Tracts.



To understand the scale of the census tracts, I also showed where the redlined neighborhoods are in comparison to the rest of Marcopa County.

4.3.1 Plotting Asthma

As shown below, asthma prevalence has a slight gradual increase correlating with the HOLC ratings with the A neighborhood having one of the lowest asthma prevalence and the D neighborhood having the highest.

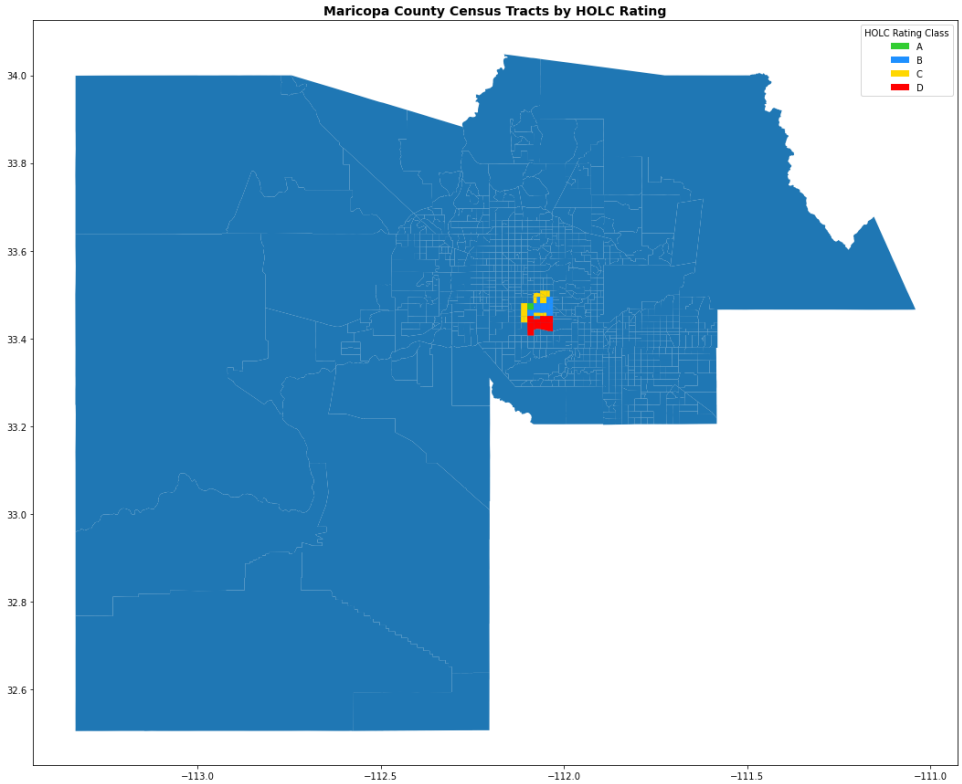
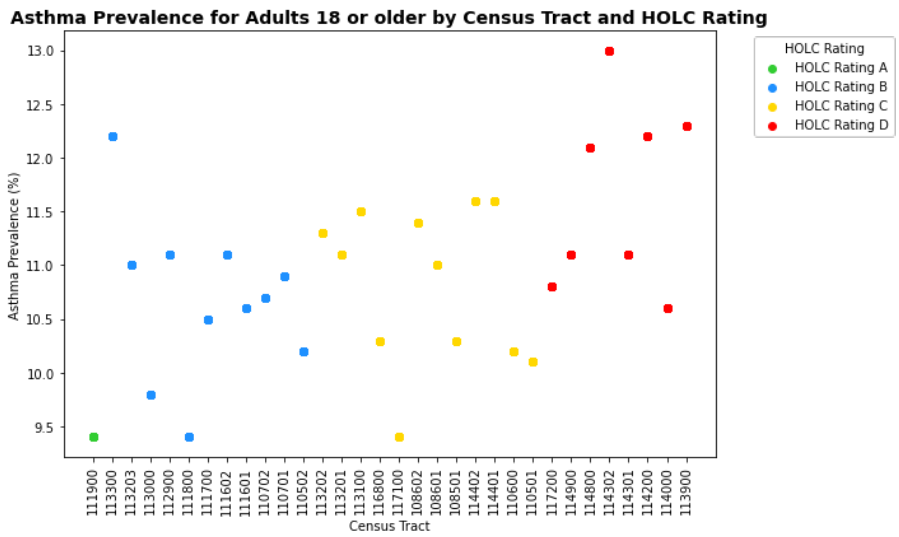
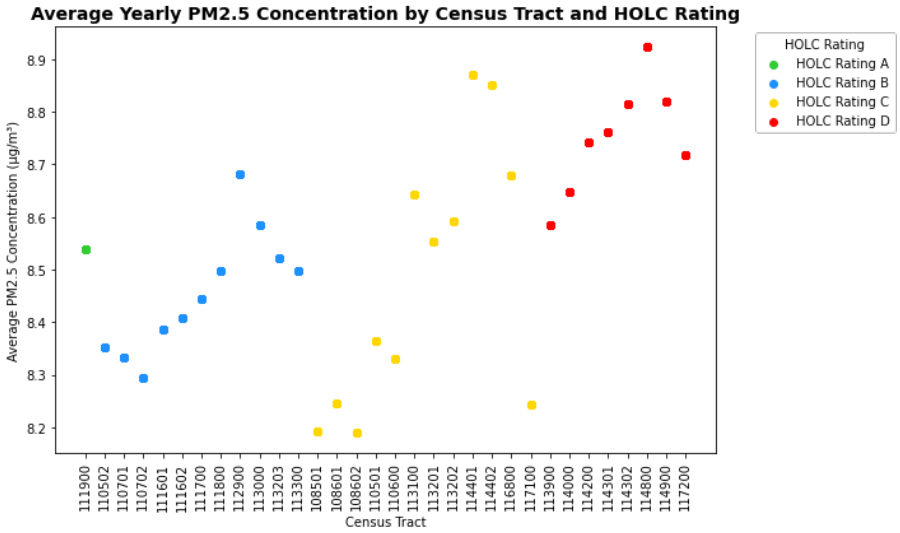


Figure 6: Enter Caption



4.3.2 Plotting air quality

The below plots depict the Average Yearly PM2.5 Concentration by Census Tract and HOLC Rating. They show that C neighborhoods have the lowest average pm2.5 concentration while D neighborhoods have the highest.


Average Yearly PM2.5 Concentration by Census Tract and HOLC Rating
Average PM2.5 Concentration (µg/m³)
8.9
8.8
8.7
8.6
8.5
8.4
8.3
8.2
111900
110502
110701
110702
111601
111602
111700
111800
112900
113000
113203
113300
108501
108601
108602
110501
110600
113100
113201
113202
114401
114402
116800
117100
113900
114000
114200
114301
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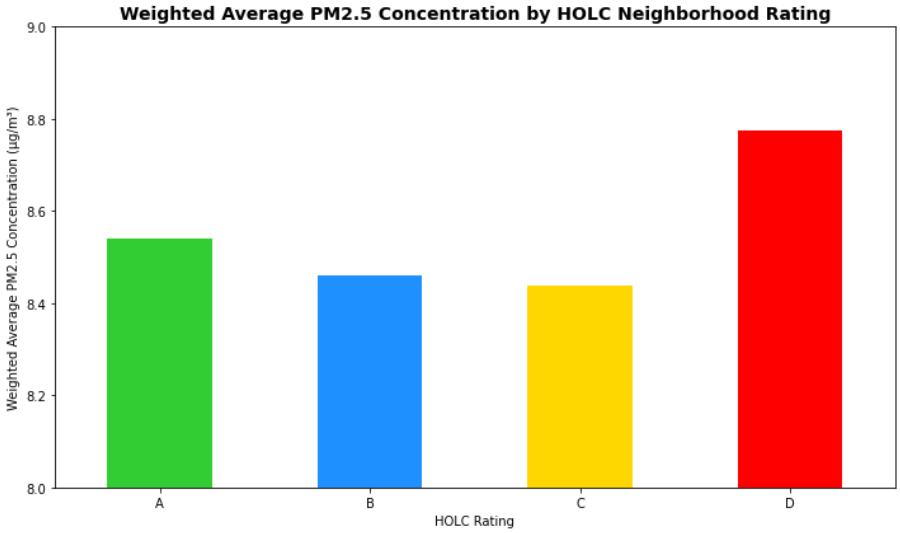
Census Tract
HOLC Rating
HOLC Rating A
HOLC Rating B
HOLC Rating C
HOLC Rating D


Weighted Average PM2.5 Concentration by HOLC Neighborhood Rating
Weighted Average PM2.5 Concentration (µg/m³)
9.0
8.8
8.6
8.4
8.2
8.0
A
B
C
D
HOLC Rating

4.3.3 Correlation Matrix

Variables used: airqualityPM25: represents air quality data related to particulate matter of size 2.5 micrometers or smaller (PM2.5). asthmaprevalence: prevalence of asthma within each HOLC district holcnumeric: a numeric representation of the HOLC (Home Owners' Loan Corporation) rating, which historically graded neighborhoods on their "desirability" with implications for investment and lending.

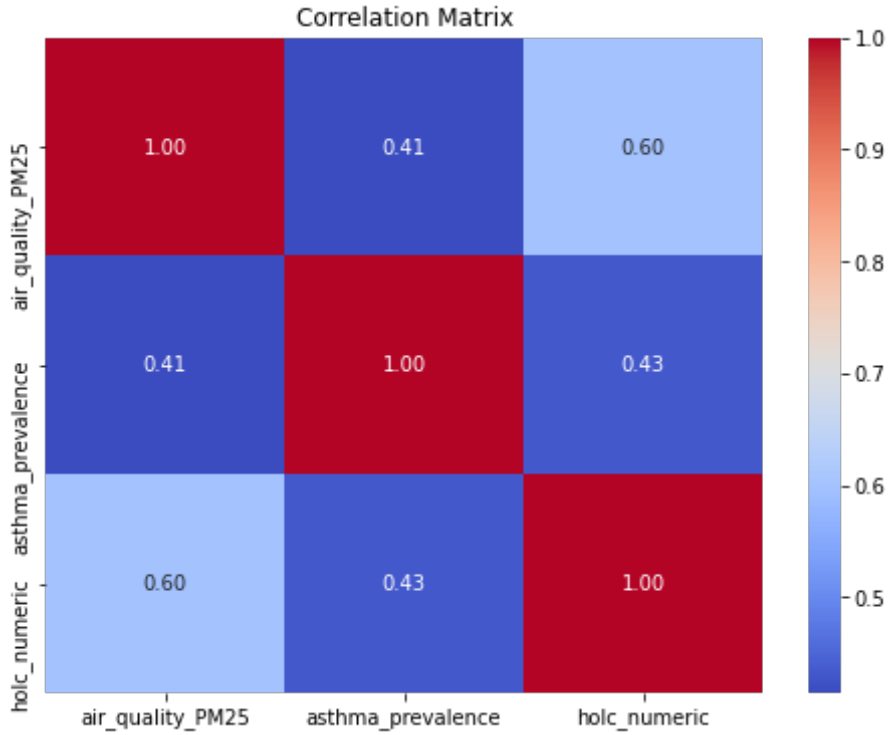
Analysis Air Quality and Asthma Prevalence: correlation coefficient of 0.41, suggesting a moderate positive correlation. As air quality worsens (higher PM2.5 levels), the prevalence of asthma also increases.

Air Quality and HOLC Rating: correlation coefficient of 0.60, which is a relatively strong positive correlation. Areas with worse HOLC ratings (higher numeric values, meaning lower desirability) have poorer air quality.

Asthma Prevalence and HOLC Rating: correlation coefficient of 0.43, there's a moderate positive correlation between these two variables. Areas with worse HOLC ratings might see a higher prevalence of asthma.

Conclusion This correlation matrix shows that lower-graded HOLC areas have a higher association with poor air quality and higher asthma prevalence.

This analysis delves into the socioeconomic characteristics associated with each HOLC grade, revealing a persistent influence of historical lending practices on present-day realities.



4.4 Income, Poverty and Educational Attainment

This analysis delves into the socioeconomic characteristics associated with each HOLC grade, revealing a persistent influence of historical lending practices on present-day realities. **Grade A** For Income and Poverty, grade A boasts the highest median household income and the lowest poverty rate, indicating a more affluent and economically stable environment. In education, this grade is characterized by the highest percentage of residents with a bachelor's degree or higher and a low percentage with incomplete high school education or no schooling completed. This highlights a well-educated population. **Grade B** For Income and Poverty, while exhibiting a slightly lower median income compared to Grade A, Grade B still maintains a relatively high income level. The poverty rate is also the highest in this Grade, indicating a shift towards a less affluent economic landscape. For education, it was similar to Grade A, a significant portion of the population holds a bachelor's degree or higher, but it is smaller. The percentage of residents with only a high school education is slightly higher than Grade A, but lower educational attainment remains uncommon. **Grade C** For Income and Poverty, the median household income dips in Grade C, making it the lowest. The poverty rate remains similar to Grade B. In education, the percentage of residents with a bachelor's degree or higher is also similar to Grade B. As well that the dominant category is holding a high school diploma or equivalent. **Grade D** For Income and Poverty, Grade D is not the lowest median household income among all grades, which might seem counterintuitive. However, this could be due to a larger disparity in income distribution, with some residents having very high incomes while others struggle financially. The poverty rate in Grade D is close to Grades B and C. As for education, while High School or Equivalent education remains dominant, the proportion with a Bachelor's or Higher degree is surprisingly higher than in Grades B and C. However, the Incomplete High School or Lower group is still present

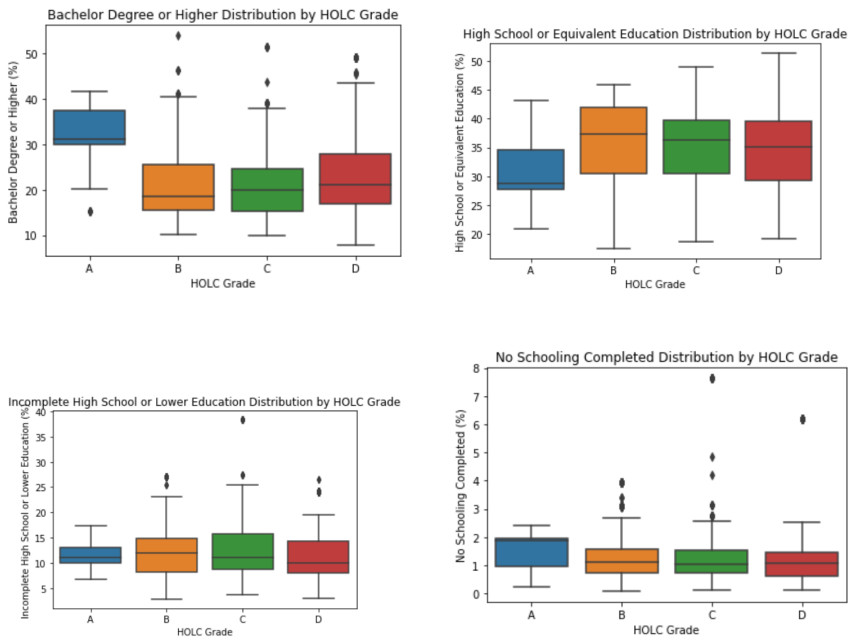


Figure 7: Educational Attainment by HOLC Grade

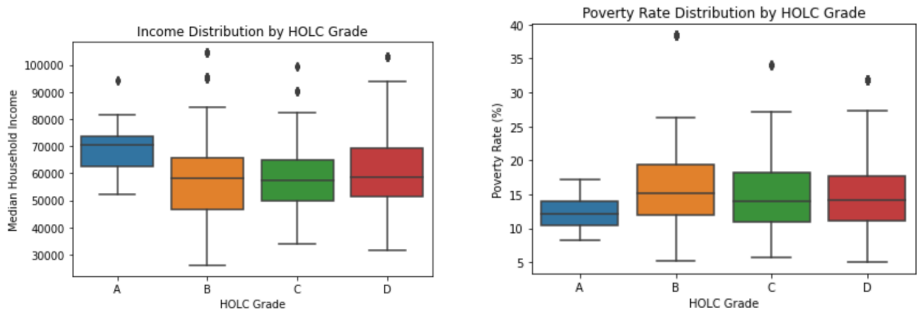


Figure 8: Economic Indicators by HOLC Grade

4.4.1 Regression Model

Including Income and Grades This model utilizes a set of predictors including median income, employment-related variables, various educational attainments, and the numerical representation of HOLC grades. The strong performance of this model, as indicated by an RMSE of 1.003 and an R^2 of 0.970, suggests a high level of accuracy in predicting poverty rates. The high R^2 value particularly highlights that the model explains a significant portion of the variance in poverty rates, which can be largely attributed to the inclusion of income and grade factors. This underscores the strong relationship between economic conditions, historical grading, and poverty.
Regression RMSE: 1.0034829436393615
Regression R^2: 0.9698588710239728

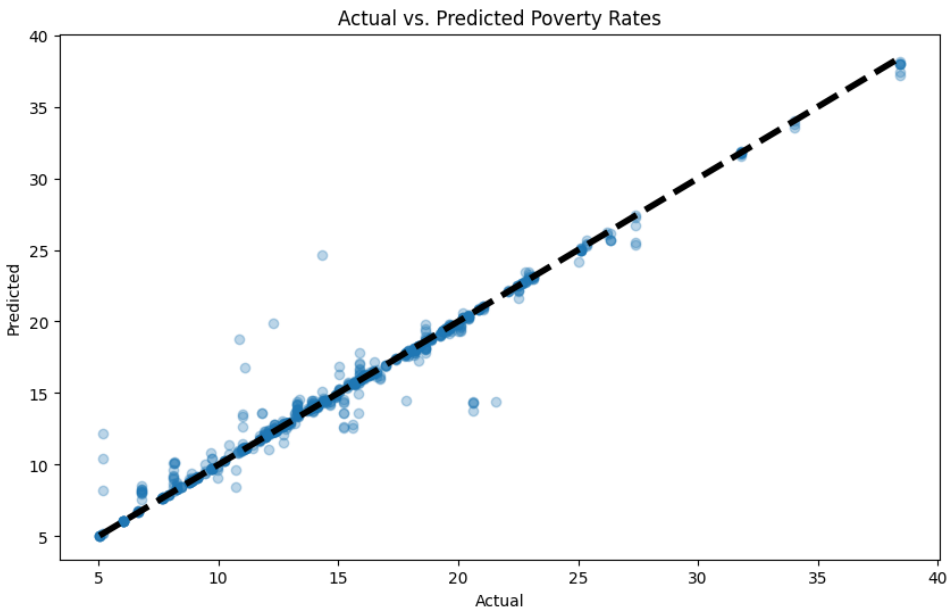


Figure 9: Regression Analysis Including Income

Excluding Income and Grades In contrast, the second model excludes income data and relies solely on educational variables and HOLC grades. The results show a considerable decrease in performance with an RMSE of 5.994 and an R^2 of -0.075. This negative R^2 value indicates that the model fails to predict poverty rates accurately and performs worse than a simple model using the mean of the dependent variable for prediction. This significant drop in model performance without income data suggests that while education and grading are important, income levels play a critical role in determining poverty rates, and their exclusion leads to inadequate model predictions.

Regression RMSE: 5.993615211915679
Regression R^2: -0.07526928657724619

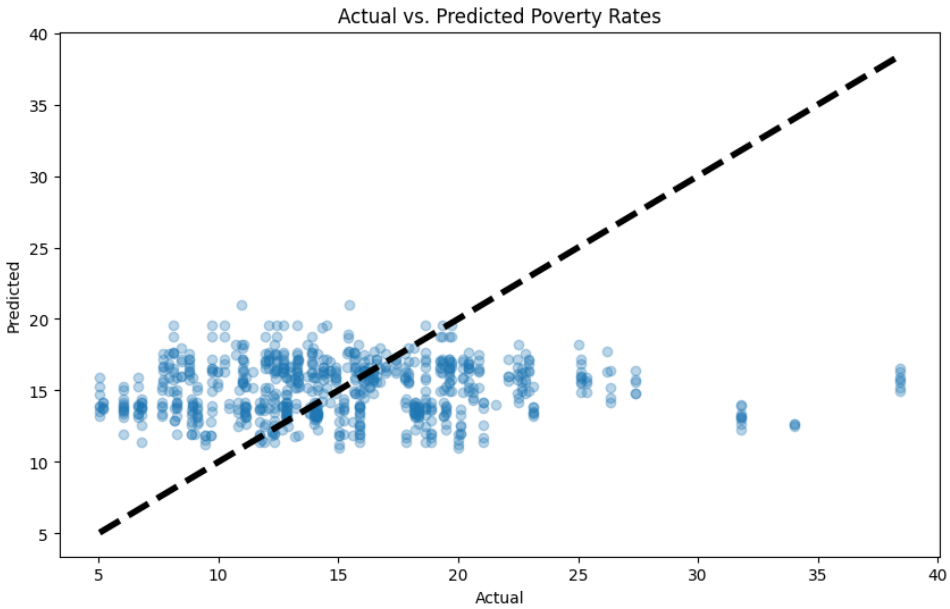


Figure 10: Regression Analysis Excluding Income

4.5 Analysis of Variance (ANOVA) Results Across HOLC Grades

The ANOVA results provide a detailed look at how historical redlining has contributed to persistent socioeconomic disparities across neighborhoods. The analysis shows significant variance in median incomes and poverty rates across HOLC grades. Areas with higher grades, deemed low-risk historically, tend to have higher median incomes and lower poverty rates. Conversely, lower-graded, high-risk areas suffer from higher poverty, reflecting long-term economic disadvantages stemming from limited access to credit and investment.The stark differences in educational attainment suggest that higher-graded areas also enjoy better educational outcomes. This educational divide parallels findings in income inequality, as indicated by the Gini Index, with lower-graded areas experiencing greater income disparity. This suggests a concentration of wealth in fewer hands in these neighborhoods, exacerbating social and economic challenges.

Table 1: ANOVA Results for Socioeconomic Indicators Across HOLC Grades

Variable	**F-value**	**p-value**
Median Income	14.773	1.557×10^{-9}
Poverty Rate	7.071	9.908×10^{-5}
Bachelor or Higher Education	32.126	2.285×10^{-20}
Gini Index	6.721	1.63×10^{-4}

4.6 Comparison of Socioeconomic Disparities Across HOLC Grades

The analysis using Tukey's Honestly Significant Difference (HSD) test effectively highlights specific socioeconomic disparities across HOLC grades. The significant differences between the highest-rated area (grade A) and lower grades (B, C, D) indicate less income inequality in grade A areas. This suggests that areas viewed historically as lower risk still enjoy more equitable income distribution. The differences between grades B and D emphasize varying levels of inequality, pointing to the lingering effects of economic policies influenced by redlining. There are pronounced income disparities, with grade A showing significantly higher median incomes compared to lower grades. This disparity is not just a marker of past injustices but continues to affect these communities' economic realities. The subtle differences between grades B and C versus the more pronounced disparity with grade D highlight how even slight variations in historical grading can lead to noticeable economic gaps.

Table 2: Multiple Comparison of Means - Tukey HSD for Gini Index

Group1	**Group2**	**Mean Diff**	**p-adj**	**Lower**	**Upper**	**Reject**
A	B	0.0086	0.0029	0.0022	0.015	True
A	C	0.0088	0.0025	0.0024	0.0153	True
A	D	0.0067	0.0353	0.0003	0.013	True
B	C	0.0002	0.9909	-0.0019	0.0024	False
B	D	-0.0019	0.0374	-0.0037	-0.0001	True
C	D	-0.0022	0.0485	-0.0043	0.0000	True

Table 3: Multiple Comparison of Means - Tukey HSD for Median Income

Group1	**Group2**	**Mean Diff**	**p-adj**	**Lower**	**Upper**	**Reject**
A	B	-11090.6022	0.0	-16842.2286	-5338.9758	True
A	C	-12760.1516	0.0	-18599.9138	-6920.3894	True
A	D	-9402.1915	0.0002	-15159.3799	-3645.0032	True
B	C	-1669.5494	0.1159	-3595.7059	256.607	False
B	D	1688.4107	0.0443	29.3526	3347.4688	True
C	D	3357.9601	0.0001	1415.2585	5300.6617	True

4.7 Correlation Matrix

The heatmap visualizes correlations between different socio-economic indicators. High positive or negative values indicate strong relationships, which can be pivotal for understanding underlying patterns and for feature engineering in predictive modeling. Observations might reveal expected correlations, such as higher education levels correlating negatively with poverty rates, or surprising insights, such as particular racial demographics correlating strongly with specific HOLC grades. The correlation matrix analysis shows relationships between socioeconomic factors in the dataset. A positive correlation was found between poverty rates and the Gini Index, suggesting that areas with higher poverty also face greater income inequality. Educational attainment showed a negative correlation with poverty, highlighting education's potential role in economic stability. Racial demographics displayed positive correlations within groups, possibly indicating concentrated communities of similar racial or ethnic identities.

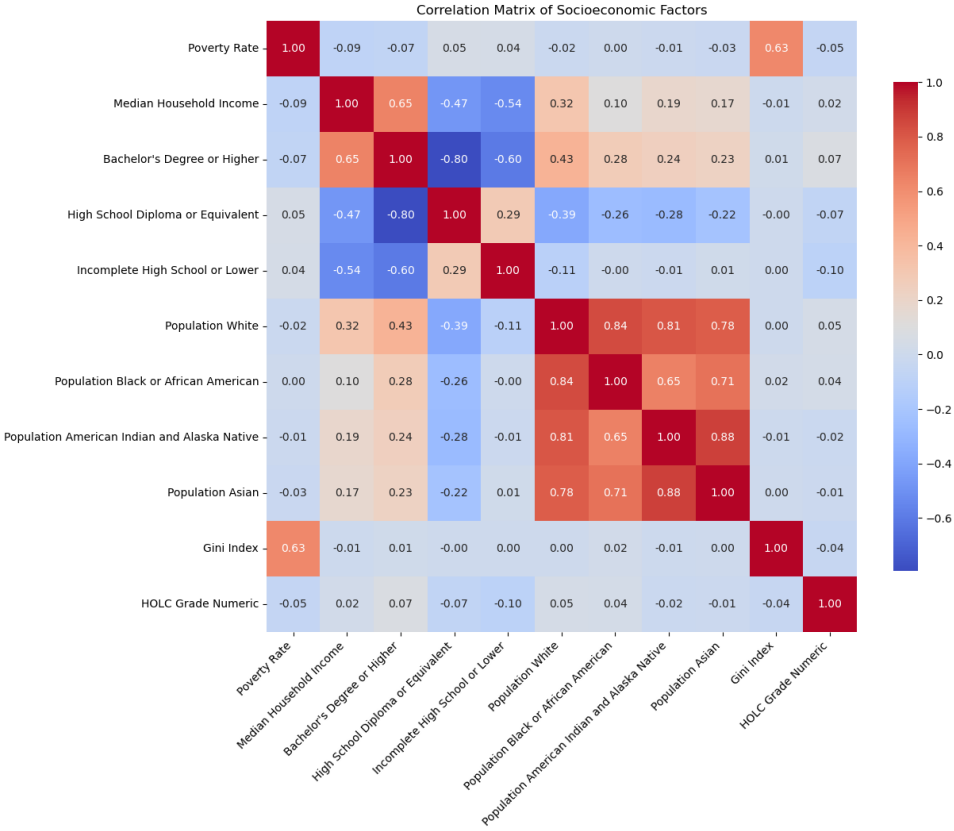


Figure 11: Correlation Matrix of Socioeconomic Factors

4.8 Random Forest Classifier Model to Predict HOLC Grades

Using a comprehensive dataset that combines population and income data from the Census and air quality data from the CDC, we trained a random forest classifier model that predicts an area's HOLC rating. Our resulting model had an accuracy of 0.9657027572293208.

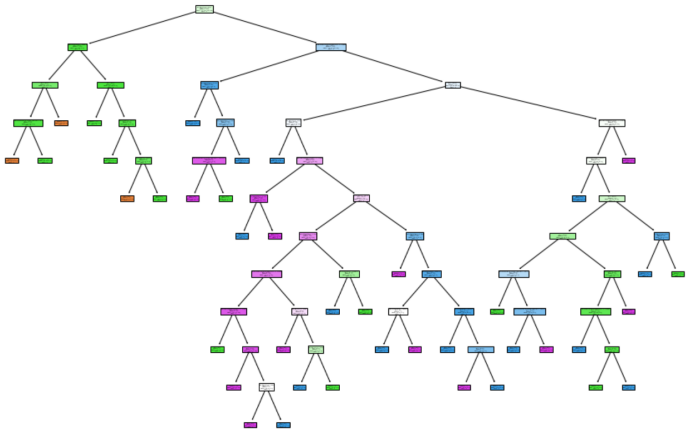


Figure 12: Decision Trees

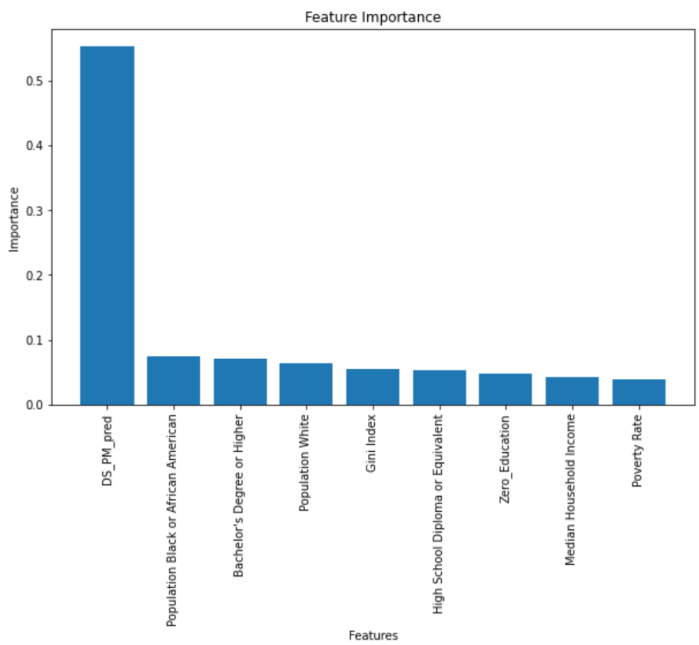


Figure 13: Feature Importance

```
Accuracy: 0.9657027572293208
              precision    recall  f1-score   support

           1       0.92      0.73      0.81        33
           2       0.97      0.97      0.97       774
           3       0.96      0.98      0.97       446
           4       0.97      0.96      0.96       234

    accuracy                           0.97      1487
   macro avg       0.95      0.91      0.93      1487
weighted avg       0.97      0.97      0.97      1487
```

Figure 14: Classification Report

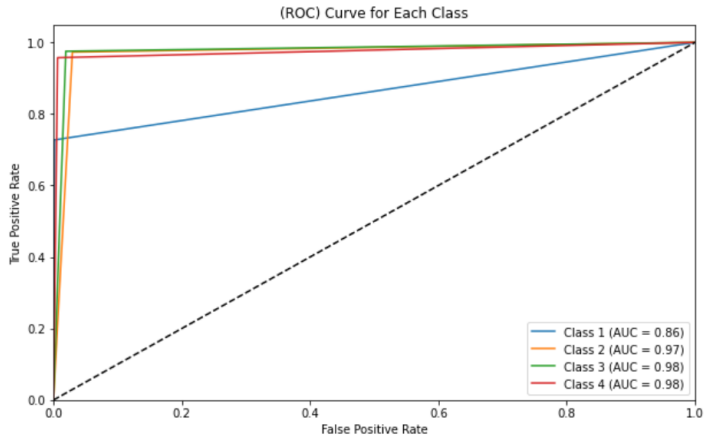


Figure 15: ROC Curve by HOLC Zone

This model reveals the concentration of PM2.5 was the biggest predictor of HOLC rating, followed by the black population of an area.

5 Discussion

The implications of this study are significant, illustrating how the remnants of redlining continue to affect communities through ongoing inequalities. These insights are crucial for developing targeted policies that address the conditions in historically marginalized areas. Furthermore, the project emphasizes the importance of comprehensive urban planning that incorporates historical data to better tackle the root causes of inequality. While our findings are robust, the limitations noted warrant cautious interpretation and suggest a need for further research that could provide a deeper understanding and more detailed analysis of redlining's impacts.

Our research faced several challenges that may have impacted the results. The geographic scope was limited to areas originally mapped by HOLC, which only cover parts of cities and have undergone considerable changes. Additionally, difficulties in integrating key data like district finance statistics and asthma prevalence into our predictive models may have limited their effectiveness. These challenges highlight the complexities of merging diverse datasets and the risks of making broad generalizations from our findings.

Overall, our findings highlight the necessity for policies that target the roots of disparities in certain neighborhoods compared to others to provide equitable development for all. Our

predictive modeling also reveals the potential for future data science techniques that assess and predict community outcomes based on historical context.

6 Conclusion

This project explored the long-term effects of redlining, across the United States. By analyzing historical data, employing statistical techniques, and using machine learning models, we documented persistent racial, socioeconomic, and environmental disparities linked to historical redlining practices. Our findings indicate a clear correlation between historically redlined areas and challenges such as lower standards of living, diminished educational outcomes, poorer health conditions, and significant economic disadvantages.

As urban areas continue to evolve, the insights provided by historical analyses like ours are crucial in paving the way for more equitable urban development.

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7 Appendix A: Code, Data Files, and Final Presentation

All files that were used in the making of this project can found at this GitHub repository: https://github.com/byslavam/Data-Science-Capstone.